

VIREN VELACHERI

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EDUCATION

Carnegie Mellon University

Master of Science in Machine Learning

Relevant Coursework: Intermediate Deep Learning, Introduction to Machine Learning, Probability and Mathematical Statistics

Pittsburgh, PA

December 2027

The University of Texas at Austin

Bachelor of Science in Computer Science with Honors, Summa Cum Laude

GPA: 3.96/4.0 | College Scholar

Relevant Coursework: Neural Networks, Natural Language Processing, Computer Vision, Data Mining, Network Security

Austin, TX

May 2022

WORK EXPERIENCE

Paloa (paloa.ai)

Founding Engineer (Part-Time)

Austin, TX

February 2026 – Present

- Built a VLM-first basketball video-analysis pipeline (Twelve Labs Pegasus, structured outputs, scoreboard corroboration) on Modal workers, producing player identification, box scores, and AI game summaries.
- Designed a layered player-identification system combining RF-DETR detection, ByteTrack tracking, jersey-OCR VLMs with majority-vote consensus, and SigLIP embedding clustering for team assignment.
- Created Paloa BENCH, a regression framework scoring outputs against 469 ground-truth shots across 4 games on five dimensions with pass/fail regression gates.
- Built a QLoRA fine-tuning and evaluation pipeline for shot detection (Qwen-VL, Modal A100 GPUs) with a human-in-the-loop workflow capturing coach corrections.

Cvent, Inc.

Software Engineer 2

Austin, TX

August 2024 – July 2026

- Architected a cross-service optimization cutting system latency by 35% and incident recovery time by 50%+ via Datadog.
- Doubled team productivity by integrating AI development frameworks into engineering workflows.
- Selected by management to join a two-person, principal engineer-led team for a feature initiative impacting 10,000+ users.

Software Engineer 1

August 2022 – August 2024

- Engineered React and Redux components that improved dashboard responsiveness by 15%.
- Designed RESTful APIs serving 1,200+ clients with WebSocket handling for 500+ messages per second.

RESEARCH PROJECTS

RAG Retrieval Benchmarking & NER Model Evaluation

Independent Research with Dr. Anshumali Shrivastava, ThirdAI, Inc.

December 2024 – July 2025

- Designed single- and multi-document RAG retrieval benchmarks with custom datasets to standardize retrieval evaluation.
- Evaluated 8 transformer NER models on precision, recall, F1 score, and sub-10-millisecond latency to guide model selection.
- Self-hosted vision-language models for document processing and authored reports informing architecture decisions.

Network Optimization for 5G & Starlink Deployments

Independent Research with Prof. Chandrajit Bajaj, The University of Texas at Austin

February 2024 – July 2024

- Analyzed dynamic network optimization using genetic algorithms, deep reinforcement learning, and graph neural networks.
- Proposed a hybrid optimization framework for 5G and Starlink network deployments based on comparative findings.

ENTREPRENEURSHIP & PROJECTS

SiliconMCP (siliconmcp.com), formerly LoopCell

Co-Founder & CEO

Austin, TX

July 2025 – Present

- Building MCP-based infrastructure for AI-driven chip design: evaluation-as-a-service for LLM-generated SystemVerilog, agentic RTL and testbench generation pipelines, and an active pilot with a leading frontier model company.
- Placed in the top 10% of Y Combinator applicants.

Resume-Bot (resume.llmchat.ai) & HireBot (hire.llmchat.ai)

Founder

March 2024 – Present

- Built a provisional-patent LLM system that turns resumes into interactive AI chatbots (Python, Streamlit, MongoDB, AWS).
- Developed an AI hiring tool that parses resumes and ranks candidates against job descriptions.

SKILLS

ML/AI: PyTorch, TensorFlow, Hugging Face, LoRA/QLoRA, YOLO, scikit-learn, Computer Vision, NLP, RAG, LLMs, VLMs

Programming Languages: Python, Java, JavaScript/TypeScript, SQL, C/C++, R, MATLAB, SystemVerilog

Frameworks & Tools: React, Node.js, Spring Boot, Django, Docker, AWS, Modal, MongoDB, PostgreSQL, Spark, Hadoop, Git

HONORS

Cvent “Pat on the Back” Award (2024) | President’s Award for Global Learning Finalist (2020)

MIT Zero Robotics World Championships, 3rd Place (2018)